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DBA361

PROJECT

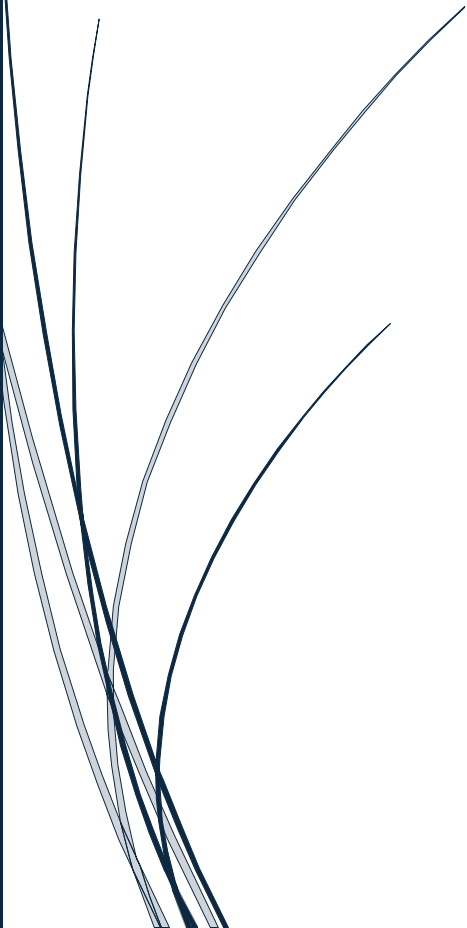
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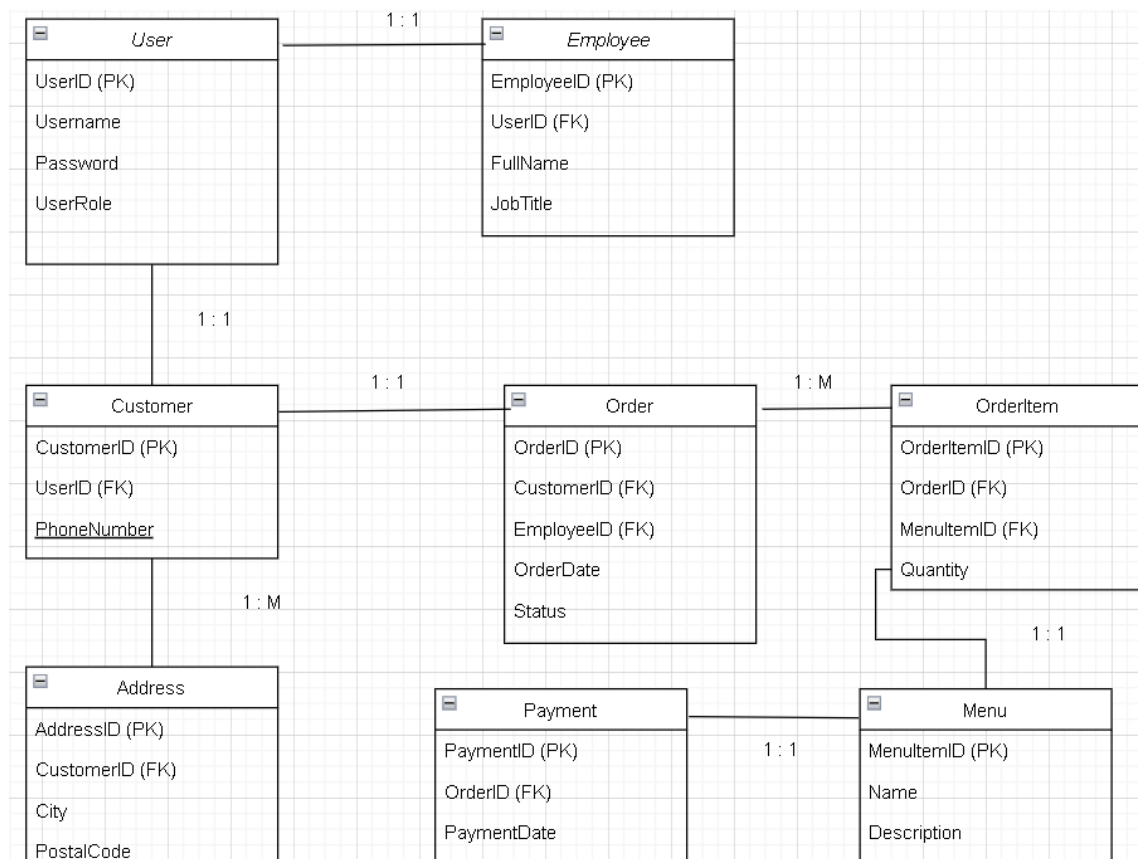
1. Introduction

This document serves as the Database Reference Manual for the Macy's Pizza ordering system. The purpose of this manual is to provide database administrators (DBAs), developers, and users with the technical details relevant to the database system. The database is designed to manage users, employees, customers, and food orders for Macy's Pizza, a local pizza restaurant.

2. Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) for the Macy's Pizza database is designed using Crow's Foot notation. The diagram includes the following entities and relationships:

- **Users:** Contains user information (staff and external users).
- **Customers:** Stores customer details.
- **Menu:** Contains menu items with descriptions and prices.
- **Orders:** Tracks customer orders.
- **OrderItems:** Links orders to menu items and tracks quantities.



3. Database Implementation and Configuration

The database was implemented using SQL Server. The following steps were taken to configure the database:

1. Database Creation:

- The database was created with a primary filegroup and a secondary filegroup.
- The recovery model was set to **Full Recovery**.
- Automatic shrinking of database files was enabled.
- Mixed Mode Authentication was enabled.

```
CREATE DATABASE MacysPizza
ON PRIMARY
    (NAME = MacysPrimary, FILENAME = 'C:\Program Files\Microsoft SQL Server\MSSQL16.SQLEXPRESS\MSSQL\DATA\MacysPizzaDB.mdf', SIZE = 50MB, MAXSIZE = 500MB, FILEGROWTH = 10MB),
FILEGROUP MacysSecondary
    (NAME = MacysSecondary, FILENAME = 'C:\Program Files\Microsoft SQL Server\MSSQL16.SQLEXPRESS\MSSQL\DATA\MacysPizza_log.ldf', SIZE = 50MB, MAXSIZE = 500MB, FILEGROWTH = 10MB)
LOG ON
    (NAME = MacysLog, FILENAME = 'C:\Program Files\Microsoft SQL Server\MSSQL16.SQLEXPRESS\MSSQL\DATA\MacysPizza_log.ldf', SIZE = 50MB, MAXSIZE = 500MB, FILEGROWTH = 10MB);
```

Schema Creation:

- A schema named Macys was created to organize the database objects.

```
USE MacysPizza
GO
```

```
CREATE SCHEMA Macys;
```

4. Tables, Schemas, and Indexes

The following tables were created under the Macys schema:

```
CREATE TABLE Macys.Users (
    UserID INT IDENTITY(1,1) PRIMARY KEY,
    Username NVARCHAR(50) NOT NULL UNIQUE,
    Password NVARCHAR(255) NOT NULL,
    UserRole NVARCHAR(20) CHECK (UserRole IN ('Admin', 'Staff', 'Customer')) NOT NULL,
) ON MacysSecondary;

CREATE TABLE Macys.Customers (
    CustomerID INT IDENTITY(1,1) PRIMARY KEY,
    Phone NVARCHAR(20),
) ON MacysSecondary;

CREATE TABLE Macys.Menu (
    MenuItemID INT IDENTITY(1,1) PRIMARY KEY,
    Name NVARCHAR(100) NOT NULL,
    Description NVARCHAR(255),
    Price DECIMAL(10,2) NOT NULL CHECK (Price > 0),
) ON MacysSecondary;

CREATE TABLE Macys.Orders (
    OrderID INT IDENTITY(1,1) PRIMARY KEY,
    CustomerID INT NOT NULL,
    OrderDate DATETIME DEFAULT GETDATE(),
    OrderStatus NVARCHAR(20) CHECK (OrderStatus IN ('Pending', 'Processing', 'Completed', 'Cancelled')) DEFAULT 'Pending',
    PaymentMethod NVARCHAR(50),
    FOREIGN KEY (CustomerID) REFERENCES Macys.Customers(CustomerID)
) ON MacysSecondary;

CREATE TABLE Macys.OrderItems (
    OrderItemID INT IDENTITY(1,1) PRIMARY KEY,
    OrderID INT NOT NULL,
    MenuItemID INT NOT NULL,
    Quantity INT NOT NULL CHECK (Quantity > 0),
    FOREIGN KEY (OrderID) REFERENCES Macys.Orders(OrderID) ON DELETE CASCADE,
    FOREIGN KEY (MenuItemID) REFERENCES Macys.Menu(MenuItemID)
) ON MacysSecondary;
```

Indexes:

- Indexes were created on frequently queried columns such as OrderStatus in the Orders table to improve query performance.

5. Users, Roles, and Permissions

Two database roles were created:

1. Management Role:

- Assigned to managers and supervisors.
- Granted **SELECT, INSERT, UPDATE** permissions on all views except the Users data.

```
CREATE ROLE Management;

GRANT SELECT, INSERT, UPDATE ON Macys.OrdersView TO Management;
GRANT SELECT, INSERT, UPDATE ON Macys.MenuView TO Management;
GRANT SELECT ON Macys.CustomersView TO Management;
GRANT SELECT ON Macys.UsersView TO Management;
```

2. Staff Role:

- Assigned to cashiers and shop floor assistants.
- Granted **SELECT** permissions on the MenuView and OrdersView.

```
CREATE ROLE Staff;

GRANT SELECT ON Macys.MenuView TO Staff;
GRANT SELECT ON Macys.OrdersView TO Staff;
```

6. Views and Authorization

The following views were created to simplify data access and restrict access to sensitive data:

```
CREATE VIEW Macys.OrdersView AS
SELECT o.OrderID, o.OrderStatus, c.Phone, m.Name AS MenuItem, oi.Quantity
FROM Macys.Orders o
JOIN Macys.Customers c ON o.CustomerID = c.CustomerID
JOIN Macys.OrderItems oi ON o.OrderID = oi.OrderID
JOIN Macys.Menu m ON oi.MenuItemID = m.MenuItemID;

GO
CREATE VIEW Macys.MenuView AS
SELECT MenuItemID, Name, Price FROM Macys.Menu;

GO
CREATE VIEW Macys.CustomersView AS
SELECT CustomerID, Phone FROM Macys.Customers;

GO
CREATE VIEW Macys.UsersView AS
SELECT Username, UserRole FROM Macys.Users;
```

7. Backup

The following backups were created:

1. Full Backup:

- A full backup of the database was taken.

```
--Make sure the C:\Backups\ directory exists before running the scripts.  
BACKUP DATABASE MacysPizza  
TO DISK = 'C:\Backups\MacysPizza_Full.bak'  
WITH INIT, NAME = 'Full Backup of MacysPizza';
```

2. Differential Backup:

- A differential backup was taken to capture changes since the last full backup.

```
BACKUP DATABASE MacysPizza  
TO DISK = 'C:\Backups\MacysPizza_Diff.bak'  
WITH DIFFERENTIAL, INIT, NAME = 'Differential Backup of MacysPizza';
```

3. Filegroup Backup:

- A backup of the secondary filegroup was taken.

```
BACKUP DATABASE MacysPizza  
FILEGROUP = 'MacysSecondary'  
TO DISK = 'C:\Backups\MacysPizza_Secondary.bak'  
WITH FORMAT, INIT, NAME = 'Secondary Filegroup Backup';
```

4. Transaction Log Backup:

- A transaction log backup was taken to ensure point-in-time recovery.

```
BACKUP LOG MacysPizza  
TO DISK = 'C:\Backups\MacysPizza_Log.bak'  
WITH INIT, NAME = 'Transaction Log Backup';
```

8. Performance Monitoring

The following tools were used to monitor database performance:

1. SQL Server Dynamic Management Views (DMVs):

- Used to monitor query performance and index usage.

```
DBCC SHOW_STATISTICS ('Macys.Orders', idx_OrderStatus);
```

2. SQL Profiler:

- Used to trace and analyze SQL Server events.

```
EXEC sp_spaceused;  
EXEC sp_helpdb 'MacysPizza';  
  
EXEC sp_who2;
```

9. Conclusion

This document provides a comprehensive overview of the Macy's Pizza database system, including its design, implementation, and configuration. The database is designed to be scalable, secure, and efficient, with appropriate roles, permissions, and backups in place to ensure data integrity and availability.